Review plan · for the lecturer

# Teaching decisions, language and scale: unit check

University

Time
45 min

Materials
Paper; The four lessons

## Learning goals

- Four lessons
- One sitting

## Activity: Work it

1. Answer all ten
2. Then read each explain

## Discussion

- Which lesson was the miss?

## Exit ticket

- Name the idea you will redo.

**Try it in your next lecture**
Redo the one item you missed, on paper, before the next course.

Quiz sheet · for students

# Teaching decisions, language and scale: unit check

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Name _____ Group _____ Date _____

**1** **In the reward lab the robot gets +10 at the charger, and all its other moves cost it something. Every value starts at 0. After the first trip that reaches the charger, which square is the first to have a positive value?**

- ◯ The square it left to enter the charger
- ◯ The start square, where the trip began
- ◯ Every square on the path, by the same amount

**2** **A robot rewarded for spinning keeps spinning in a puddle. The fix is to change the reward, not to give it more practice trips.**

- ◯ True
- ◯ False

**3** **Match what the reward lab shows to the word.**

| | | |
|---|---|---|
| +10 at the charger | • • • • | Value |
| How dark a square is shaded | • • • • | Policy |
| The arrow on a square | • • • • | Reward |

**4** **In the teach lab, a fine-tune with no old conversations mixed in makes the bus tool work, but many old skills now fail. What explains both?**

- ◯ The model added the village names to its vocabulary
- ◯ The top block's weights moved to fit the new examples
- ◯ The frozen lower blocks were retrained on the bus examples

**5** **A language model can state a wrong date with confidence without having looked it up anywhere.**

- ◯ True
- ◯ False

**6** **Match the teach-lab run to what its test shows.**

| | | |
|---|---|---|
| Before any teaching | • • • • | The old skills hold; each step reads four times as many examples |
| Taught with no old conversations | • • • • | It does not call the bus tool |
| Taught with 48 old conversations mixed in | • • • • | The bus tool works; many old skills fail |

**7** **A student writes: "Our model scored 96% on 50 test questions, so it is safe to deploy." What do you ask first?**

- ◯ How fast the model answered
- ◯ Whether 96% is above the 90% pass mark
- ◯ Which cases it had to refuse, and whether it refused them

**8** **A model that refuses every question can never fail the suite.**

◯ True ◯ False

**9** **Match the failure to the duty it breaks.**

| | | |
|---|---|---|
| Used before a single refusal case was tried | • • • • | A person remains responsible |
| Marks changed, and no log shows which | • • • • | Test before use |
| A harmful message went out, and nobody answers for it | • • • • | Keep actions visible |

**10** **Count as the split lab does: weights plus activations. Two machines each hold 28. A layer has 16 weights, and a batch of 2 gives 2 × 16 activations. Which split fits?**

◯ Split the layer: each machine holds 8 + 2 × 8 = 24 ◯ Neither: only a bigger machine can train it

◯ Split the examples: each machine holds 16 + 1 × 16 = 32, which fits

Answers · for the lecturer

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1 **The square it left to enter the charger**

Only the move into the charger sees the +10 at once: 0 + 0.5 × 10 = 5.

2 **True**

More trips only learn the wrong reward better.

3 **+10 at the charger — Reward**
**How dark a square is shaded — Value**
**The arrow on a square — Policy**

The world pays the reward; the shading shows the value the robot has learned for a square; the arrow is its policy there, the move it makes.

4 **The top block's weights moved to fit the new examples**

Only the top block learned. Its weights now serve the bus examples, which can crowd out old answers; the frozen blocks and the vocabulary did not change.

5 **True**

It has no database of facts to look up: it guesses the next token from patterns, and a confident guess can still be wrong.

6 **Before any teaching — It does not call the bus tool**
**Taught with no old conversations — The bus tool works; many old skills fail**
**Taught with 48 old conversations mixed in — The old skills hold; each step reads four times as many examples**

Mixing old conversations back in protects what the model knew, at the cost of a longer run; leaving them out is faster but crowds old skills out.

7 **Which cases it had to refuse, and whether it refused them**

A score on questions it may answer says nothing about the cases it must refuse.

8 **False**

Refusing ordinary questions is a failure too: the suite holds the cases the model must answer as well as the ones it must refuse.

9 **Used before a single refusal case was tried — Test before use**
**Marks changed, and no log shows which — Keep actions visible**
**A harmful message went out, and nobody answers for it — A person remains responsible**

Name the duty from the failure: what nobody tested, what nobody could see, and the harm nobody owns.

10 **Split the layer: each machine holds 8 + 2 × 8 = 24**

Splitting the examples copies all 16 weights onto each machine: 16 + 16 = 32, more than 28.